

Change Notes from Homework 4 to Homework 5

Authentication and User Account Changes

- Added support for web-based user login.
 - Added an HTML login page for the web interface.
 - Added session cookie support.
 - Valid credentials create a session cookie.
 - Session cookies expire when the browser closes.
 - Added validation for invalid web login attempts.
 - Prevented unauthorized users from accessing protected web pages.
 - Prevented the web server from starting if user accounts have not been created.
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User Interface Changes

Terminal Interface

- Expanded the terminal interface to support Homework 5 features.
- Added support for editing existing records.
- Updated the briefs menu so users can:
 - List briefs.
 - Read a brief.
 - Add a brief.
 - Edit a brief.
 - Return to the main menu.
- Updated the agents menu so users can:
 - List agents.
 - Read an agent.
 - Add an agent.

- Edit an agent.
- Return to the main menu.
- Updated the facilities menu so users can:
 - List facilities.
 - Read a facility.
 - Add a facility.
 - Edit a facility.
 - Return to the main menu.
- Added Mission support to the user interface.
 - List missions.
 - Read a mission.
 - Add a mission.
 - Edit a mission.
 - Return to the main menu.

HTML Interface

- Added a read-only HTML interface.
 - Added HTML pages for:
 - Briefs.
 - Agents.
 - Facilities.
 - Missions.
 - Audit log.
 - Added navigation between HTML pages.
 - HTML users can:
 - Browse records.
 - Read records.
 - Review the audit log.
 - HTML users cannot:
 - Create records.
 - Edit records.
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Program Control Changes

- Updated ProgramControl to support both terminal and web interfaces.

- Added support for starting an HTTP server.
 - Added shared control logic between:
 - Terminal interface.
 - HTML interface.
 - Added support for handling:
 - HTTP requests.
 - Session validation.
 - HTML responses.
 - Updated navigation so both interfaces function similarly.
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Database Changes

- Updated the database to support Homework 5 relationships.
 - Expanded Mission support.
 - Missions belong to one Facility.
 - Missions may contain zero or more Briefs.
 - Agents belong to one Facility.
 - Agents may participate in multiple Missions.
 - Agents assigned to a Mission may come from any Facility.
 - Added database methods for:
 - Assigning agents to missions.
 - Retrieving agents by mission.
 - Retrieving missions by facility.
 - Updated database methods for Homework 5 functionality.
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Model and Repository Changes

- Updated relationships between:
 - Missions.
 - Briefs.
 - Agents.
 - Facilities.
- Updated repository interfaces for Homework 5 functionality.

- Updated DatabaseManager implementations to support the expanded relationships.
 - Integrated existing Mission data into the terminal and HTML interfaces.
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HTML View Changes

- Added HTML rendering methods.
 - Wrapped displayed information in appropriate HTML tags.
 - Added HTML formatting using:
 - Headers.
 - Tables.
 - Lists.
 - Hyperlinks.
 - Reused existing terminal output where practical.
 - Added HTML pages for:
 - Login.
 - Main menu.
 - Record lists.
 - Record details.
 - Audit log.
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Documentation Changes

- Updated the UML diagram.
 - Updated the ERD diagram.
 - Added wireframe documentation for the HTML interface.
 - Updated implementation notes for Homework 5 features.
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Testing Changes

- Updated UI integration tests to reflect the new menu structure.

- Added tests for:
 - HTTP server startup.
 - HTML login.
 - Session authentication.
 - HTML page generation.
 - Terminal record editing.
 - Mission navigation.
 - Agent-to-mission assignments.
 - Invalid web login attempts.
 - Updated integration tests for:
 - Terminal mode.
 - Web mode.
 - Shared control logic.
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Build and Gradle Changes

- Confirmed the project builds with Gradle using:
 - ./gradlew clean build
 - ./gradlew run
- Confirmed the terminal version continues to function.
- Confirmed the web version launches on port 3002.
- Confirmed both interfaces use the same underlying application data and control logic.